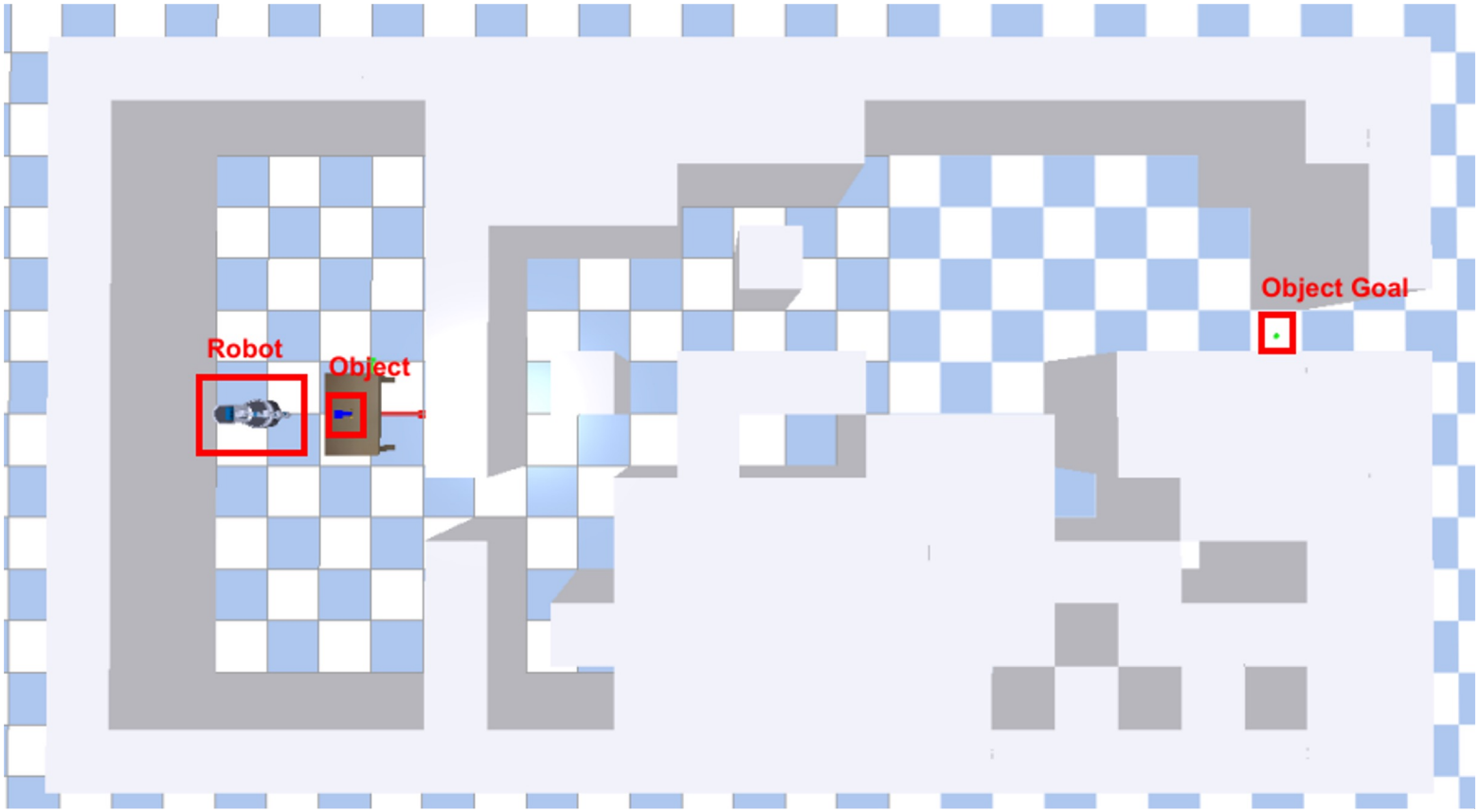
Pick-Navigation-Reach Task

Default Project for CS4278/5478

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Task Scene



Task Goal

In this project, the task is to:

1. Generate antipodal grasp proposals, solve IK, and pick up an object on the table.
2. Carry the object and navigate the robot through the maze.
3. Bring the object to the goal position (visualized as the green sphere).

Installation

1. Create a conda environment using Python 3.10.

   conda create -n pnr python==3.10

   conda activate pnr

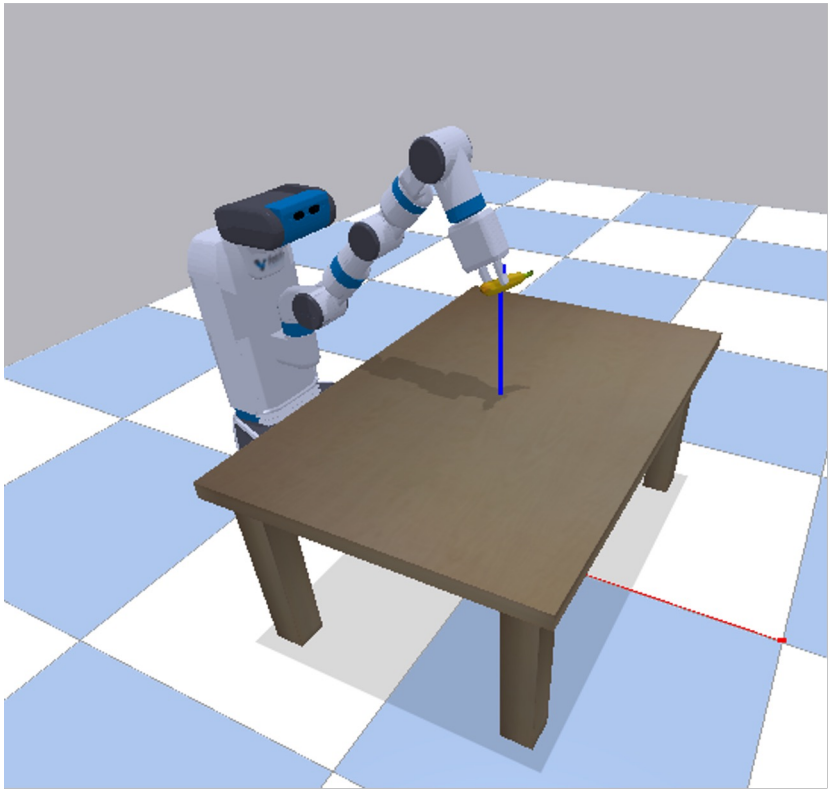
1. Our environment is build on PyBullet. Install it with pip:

   pip3 install pybullet numpy matplotlib trimesh

1. Clone the project repo:

   git clone https://github.com/NUS-LinS-Lab/Pick-Nav-Reach-Default-Project.git

Keyboard Teleoperation Demonstration





Core Function – env.step()

- **input**
  - **action** (numpy.ndarray): absolute joint value targets, (15,) # (num_dofs,)
- **output**
  - **obs** (dict):
    - **qpos**: <class 'numpy.ndarray'>, (15,) # (num_dofs,)
    - **qvel**: <class 'numpy.ndarray'>, (15,) # (num_dofs,)
    - **object_pos**: <class 'numpy.ndarray'>, (3,) # position
    - **object_xyzw**: <class 'numpy.ndarray'>, (4,) # quaternion xyzw
    - **goal_pos**: <class 'numpy.ndarray'>, (3,) # position
    - **object_pc**: <class 'trimesh.caching.TrackedArray'>, (1024, 3) # (num_points, position)
    - **object_normals**: <class 'numpy.ndarray'>, (1024, 3) # (num_points, direction)
    - **cube_positions**: <class 'numpy.ndarray'>, (140, 2) # (num_cubes, xy_position)
- **info** (dict)
  - **dist_to_goal**: <class 'numpy.float64'>, () # < 0.1 means success
  - **success**: <class 'numpy.bool'>, () # success or not

Rubics

We evaluate your algorithms in the following two aspects with 5 Unprovided seeds:

- Pick: 5 test objects (within the provided objects).
- Navigation and Reach: 5 test mazes.

Unless otherwise stated: placement tolerance = 10 cm

- **Functions implementation**: 60%
  - Grasp generation: 20%
  - Motion Planning and Grasping: 20%
  - Navigation and Reach: 20%
- **Success evaluation**
  - Pick: 24%
  - Navigation and Reach: 16%